LING2136 / 4136

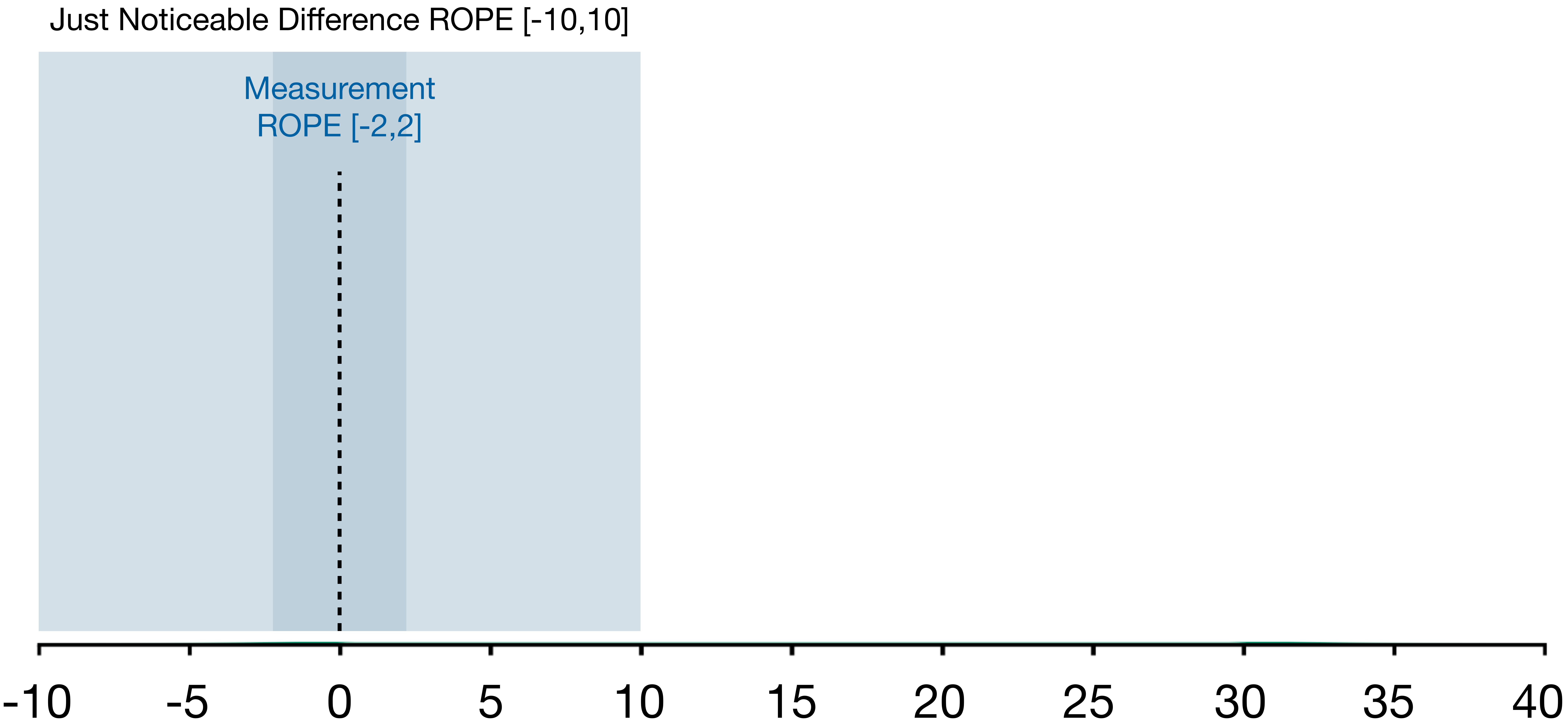
Advanced Statistical Methods for Language Students

Session-13: Wrap-Up

Lecturer: Timo Roettger

Incomplete Neutralization Experiment

n = 16 speakers x 24 items



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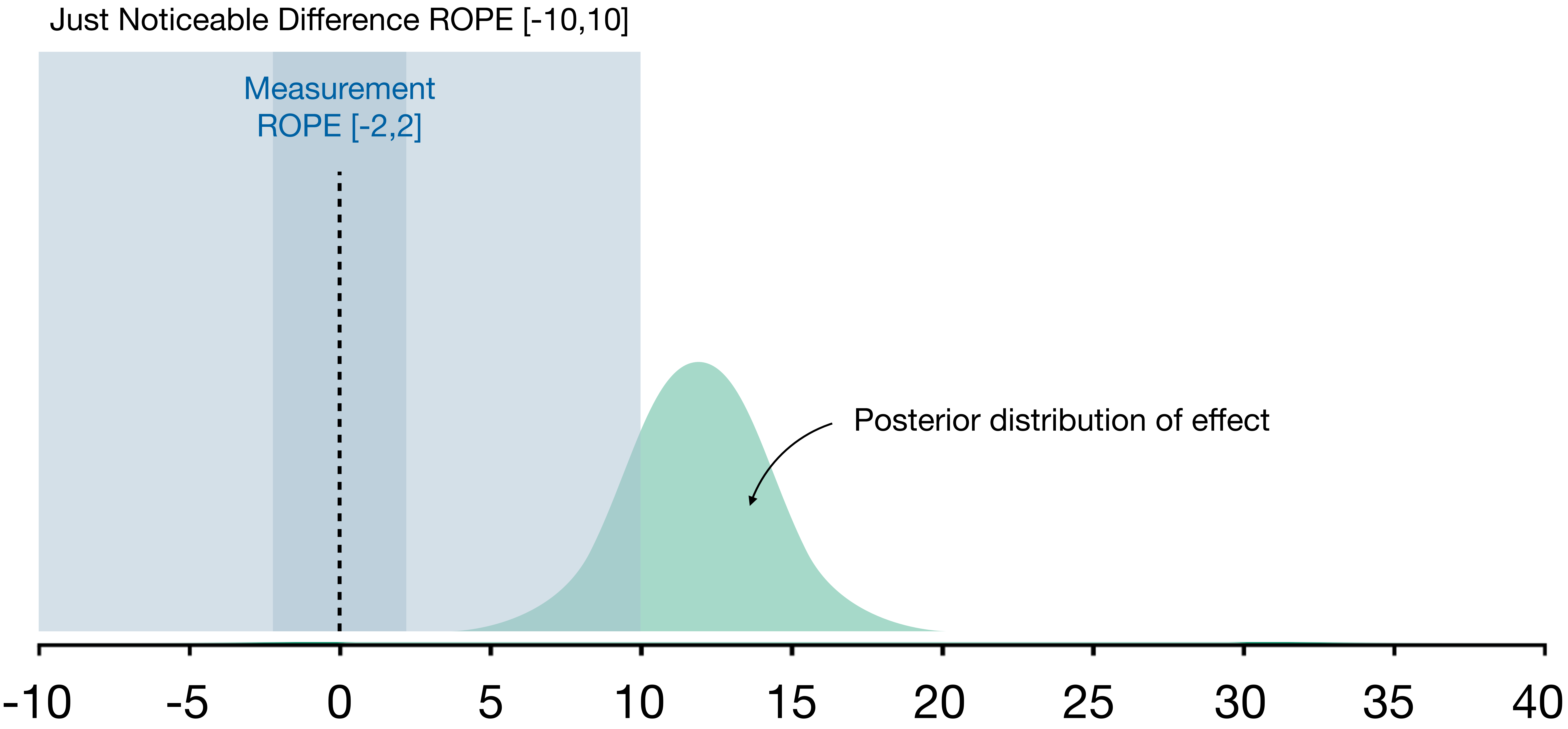

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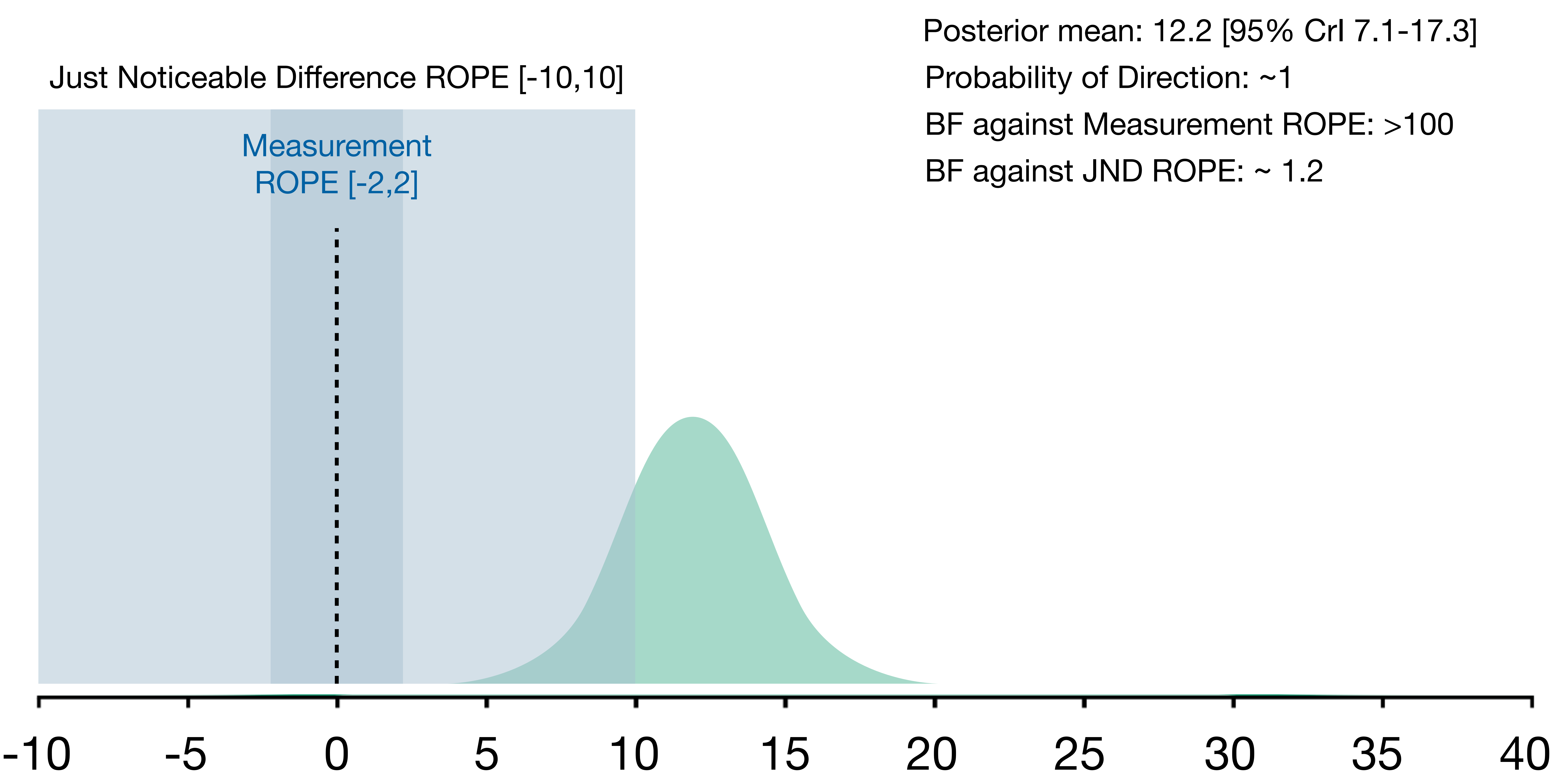
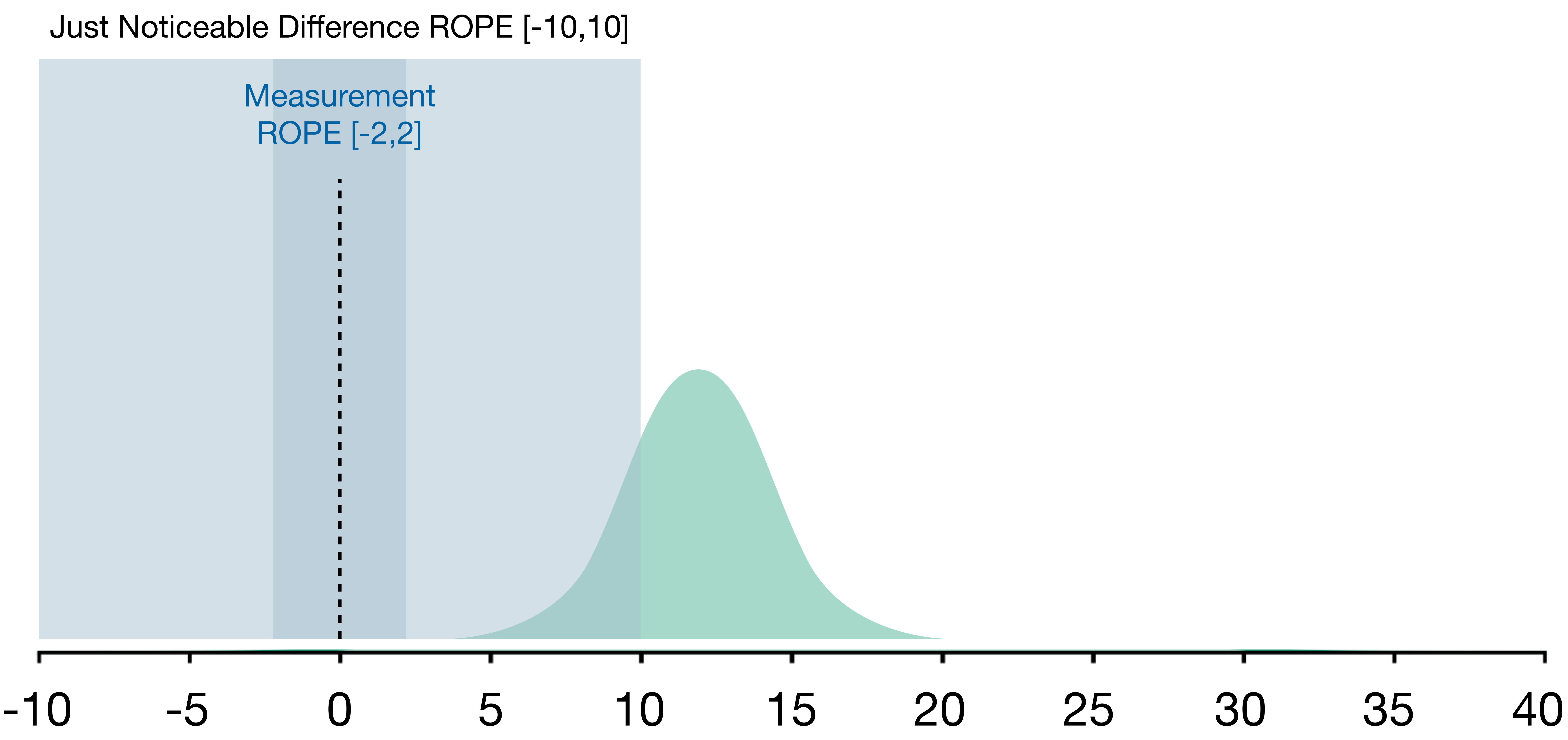


We conclude that the observed incomplete neutralization effect **is not large enough** to have communicative consequences…

Incomplete Neutralization Experiment

n = 160 speakers x 24 items



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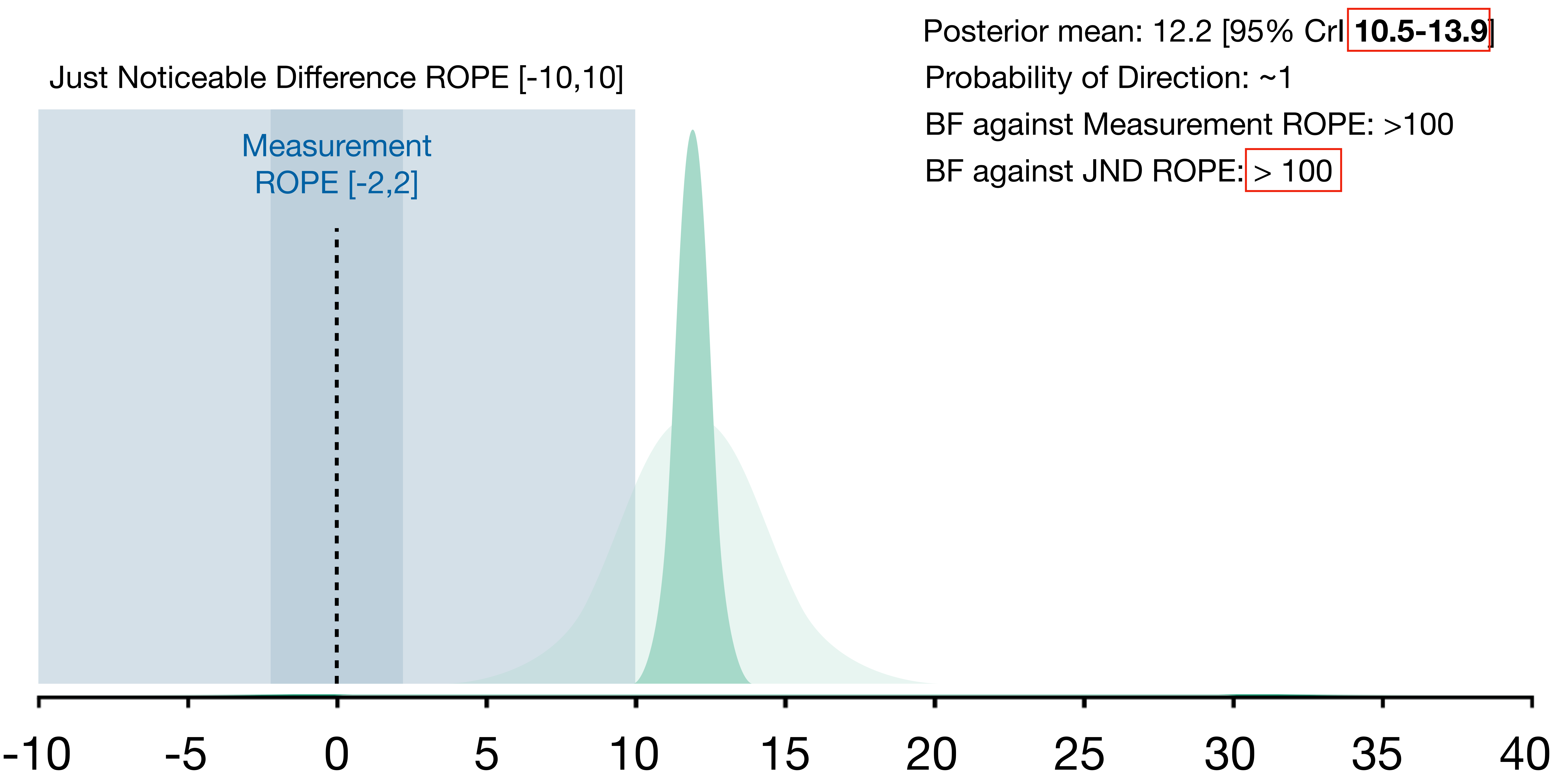


Incomplete Neutralization Experiment

n = 160 speakers x 24 items

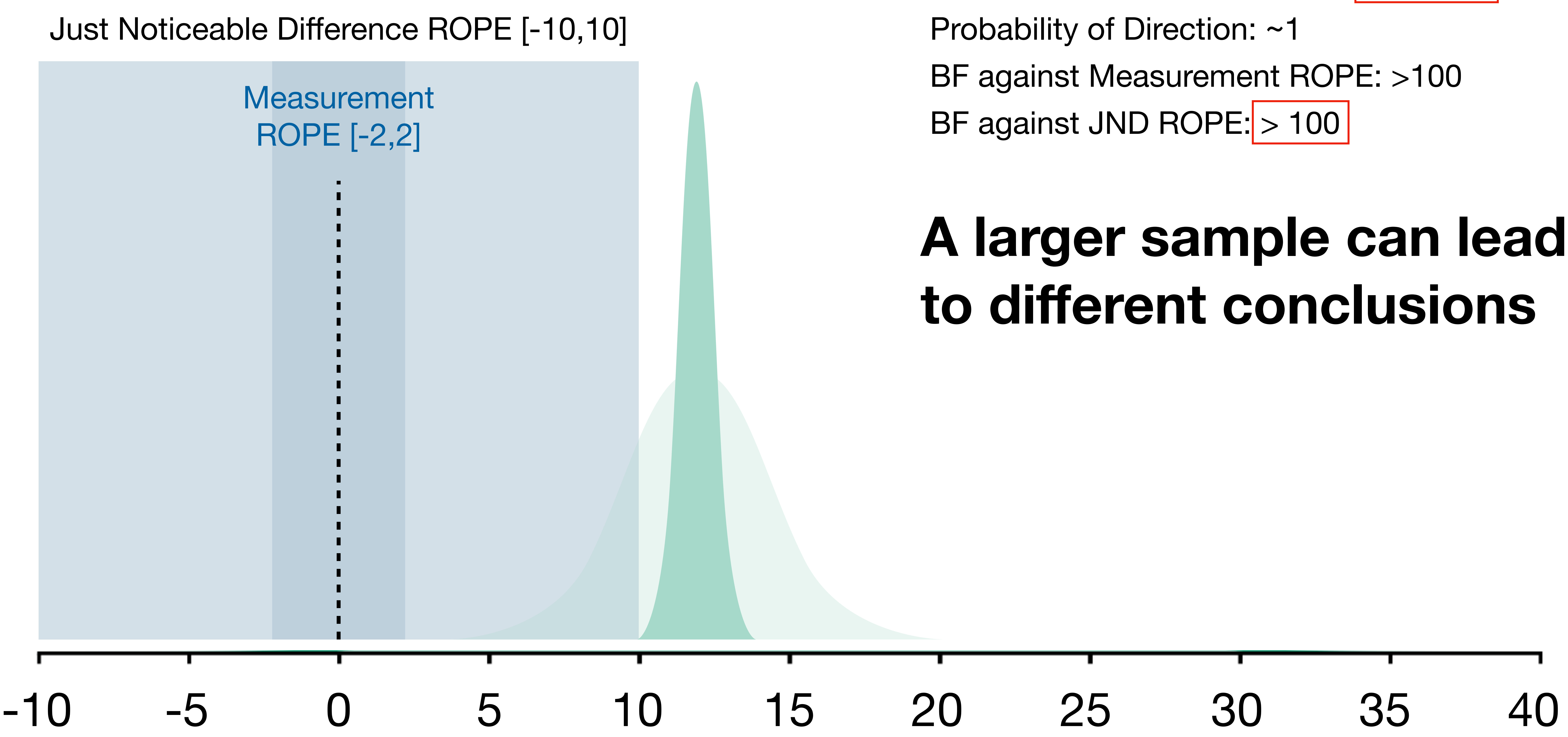
Posterior mean: 12.2 [95% CrI **10.5-13.9**]

Probability of Direction: ~1

BF against Measurement ROPE: >100

BF against JND ROPE: > 100

A larger sample can lead to different conclusions



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Given the **DATA**, the model, and our prior assumptions,…

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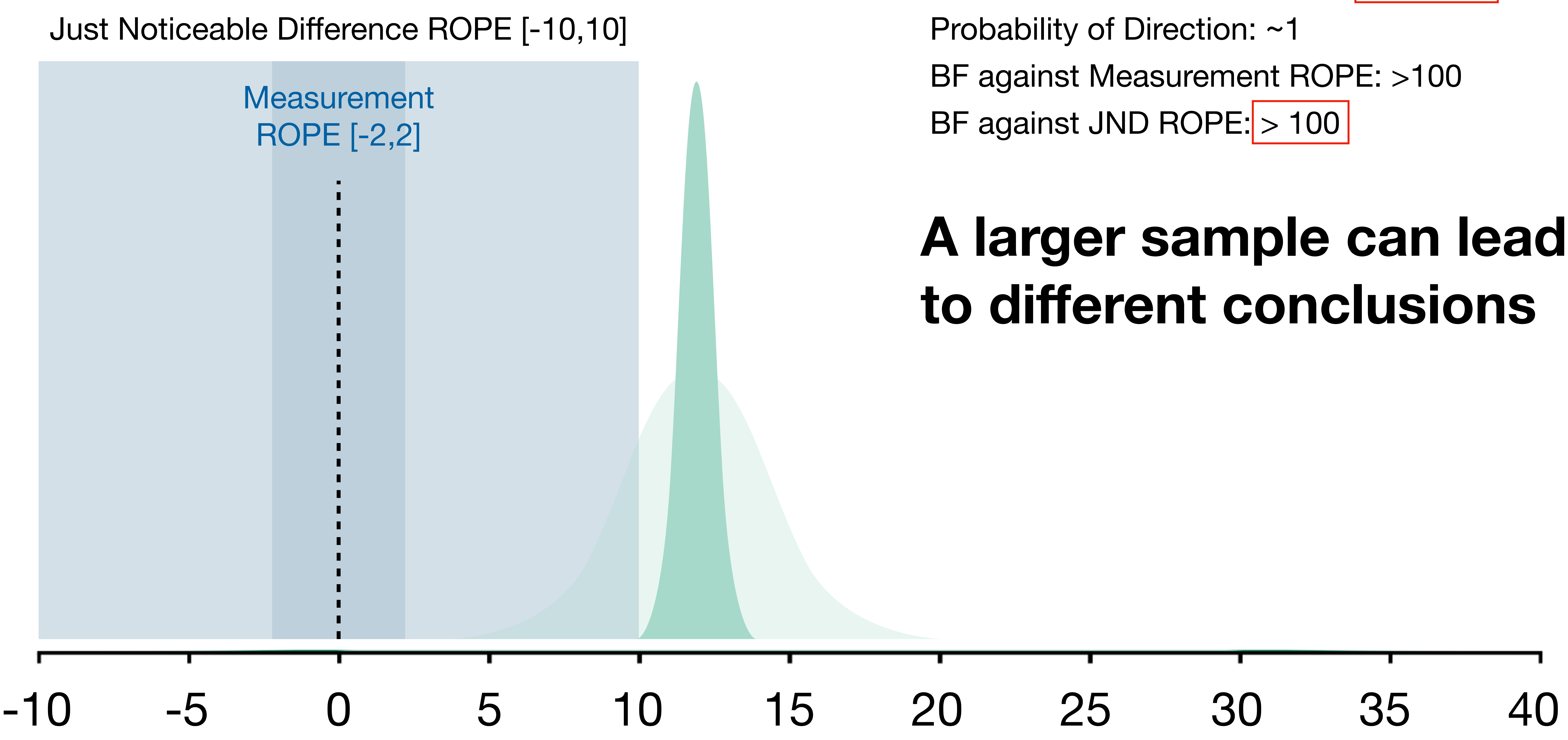

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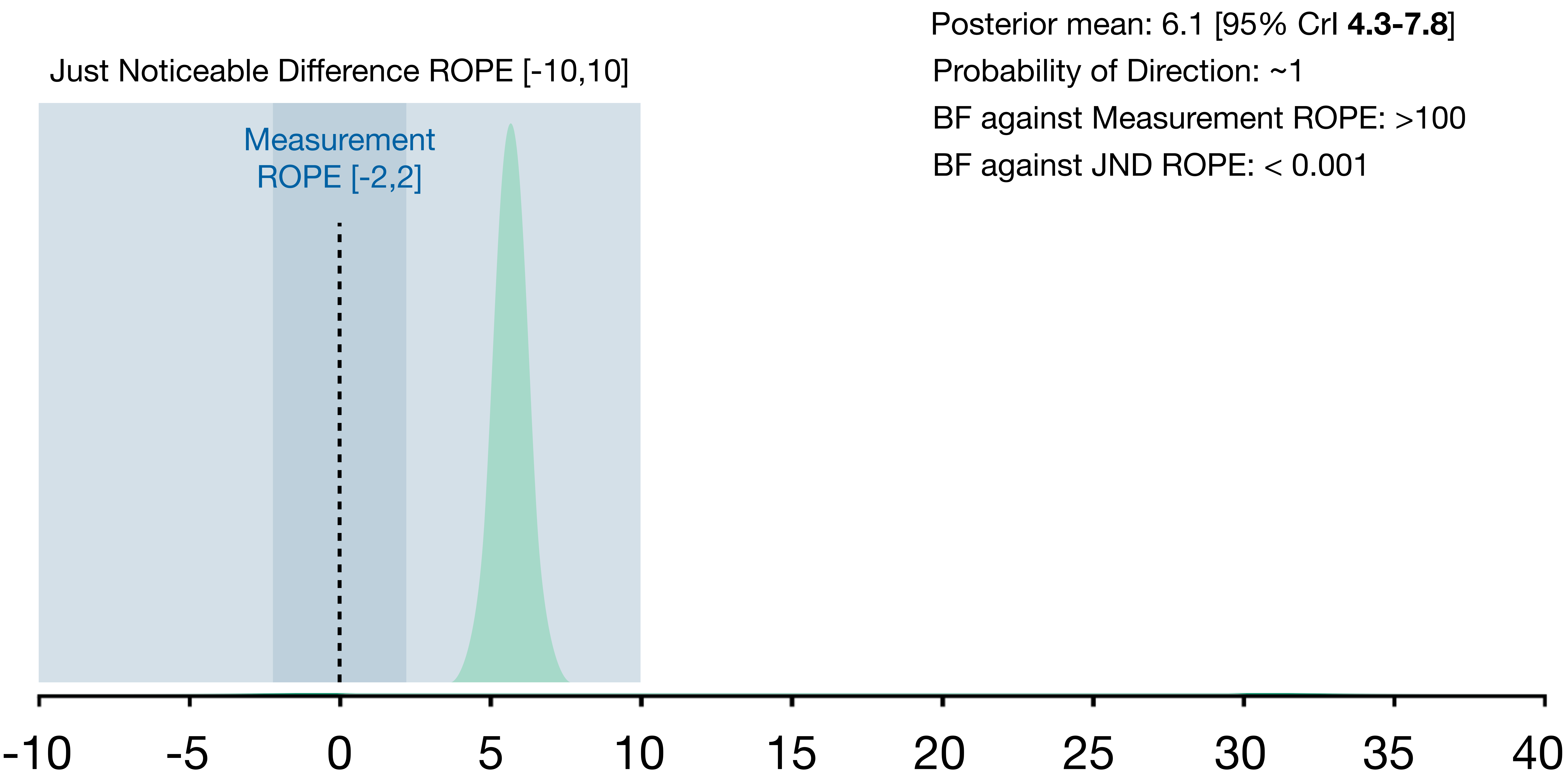

Sample Size Justifications

https://lakens.github.io/statistical_inferences/08-samplesizejustification.html

No justification	A researcher has no reason to choose a specific sample size, or does not have a clearly specified inferential goal and wants to communicate this honestly.

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Measure entire population	A researcher can specify the entire population, it is finite, and it is possible to measure (almost) every entity in the population.
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Accuracy / Sensitivity analysis

we simulate “random” datasets with specified model parameters:

Intercept

random intercepts

residual variance

DV

$$\mu_{ij} = \beta_0 + \beta_1 x_{ij} + u_{0j} + u_{1j} x_{ij} + \epsilon_{ij}$$

effect

random slopes

Expected Intercept?

Expected effect?

Expected variation between participant’s baselines?

Expected variation between participant’s effects?

Expected residual variance?

```
my_sim_data <- function(

# participants
subj_n = 50,    # number of subjects per language

# predictors
b0 = -10,        # intercept
b1 = 10,         # fixed effect

# random effects
u0s_sd = 2.7,      # random intercept SD for subjects
u1s_sd = 5.2,      # random b1 slope SD for subjects
r01s = 0.3,        # correlation between random effects b0 and b1 for subjects

sigma_sd = 4) {    # residuals SD

# create data frame
df <-   add_random(subject = subj_n) |>
        # set up within subject predictors
        add_within("subject",
                   voicing = c("voiced", "voiceless")) |>
        # set up by_subject random intercept and slope for b1
        add_ranef("subject", u0s = u0s_sd, u1s = u1s_sd, .cors = r01s) |>
        # add error term (by observation)
        add_ranef(sigma = sigma_sd) |>
        # generate responses
        mutate(response = b0 + u0s + (b1 + u1s) * delta_f + sigma)

df
}
```

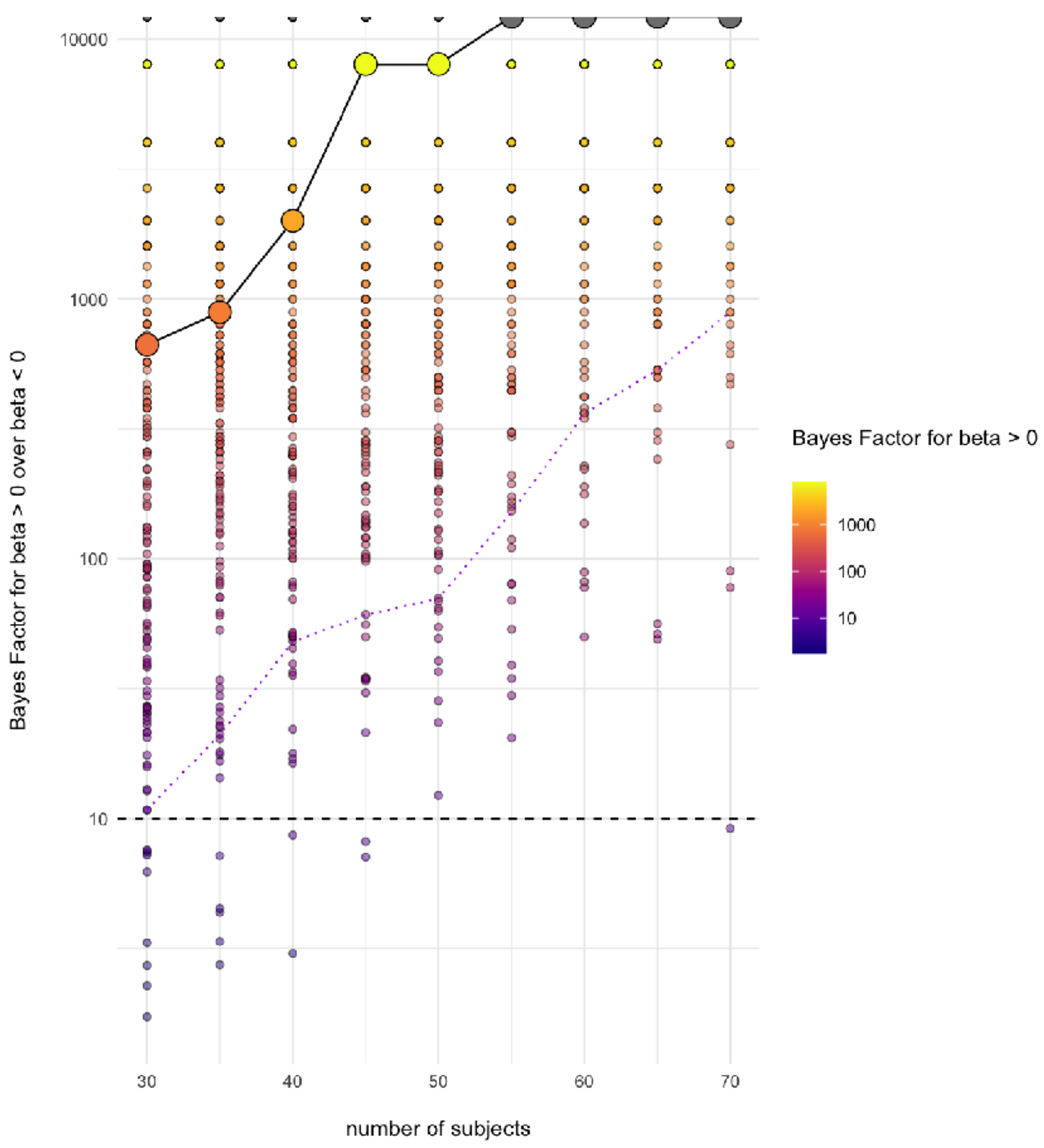
Accuracy / Sensitivity analysis

we simulate a “random” dataset with specified model parameters and different dataset sizes (e.g. participant numbers)

we run model and do inference (e.g. calculate BF)

we store the result

we repeat…


Bayes Factor for beta > 0 over beta < 0
10000
1000
100
10
30
40
50
60
70
number of subjects
Bayes Factor for beta > 0
1000
100
10

Exam

Explore, process, and analyse the data:

- Use the Bayesian framework to estimate the influence of the independent variables on the dependent variables.

- Construct and justify an appropriate (generalized) linear model.

- Specify and justify appropriate priors, describe the results with appropriate and useful visualizations.

- Critically assess the strength of evidence.

- Assess model diagnostics.

- Write up your results for experts (scientific standard)

- End with a short policy recommendation section written for non-specialist decision makers. In this section, translate your statistical findings into plain language and explain what, if anything, governments should currently do, monitor, or avoid doing on the basis of the evidence. Your recommendation should be cautious, evidence-based, and proportionate to the strength and practical relevance of your results. Avoid unexplained statistical jargon in this final section.

Use cutting-edge methods carefully

in your MA thesis / publications

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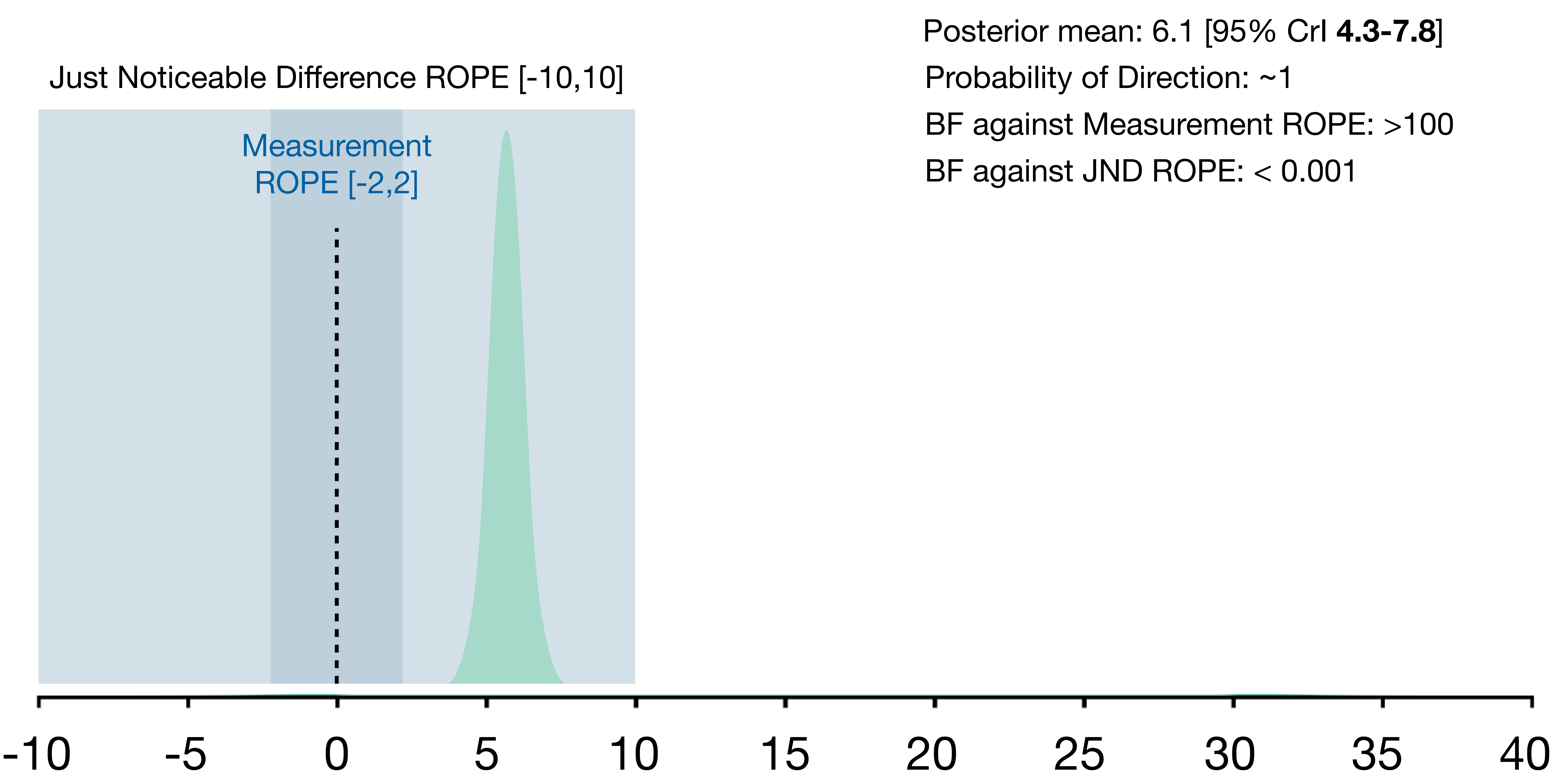


I can't
do it

EXPLICIT GOALS FOR THIS CLASS

learn all relevant aspects of **hypothesis testing** using R for **quantitative researchers** in **2026**

advance your knowledge using the R programming language	pretty common
run and interpret **hierarchical linear models**	pretty common
reproduce published findings using the original code	rare
plan and create highly informative **data visualizations**	rare
make inference using the **Bayesian framework**, including	
• specifying **prior** assumptions	rare
• defining **smallest effect sizes of interest (SESOI)**	VERY rare
• testing hypotheses using **Bayes factor**	VERY rare
work, collaborate, and store data in reproducible ways using **GitHub**.	rare


you in week 2
you in week 6
THIS CLASS WAS AN
EMOTIONAL ROLLERCOASTER